

4.2 Alcohols, haloalkanes and analysis

This section introduces two further functional groups: alcohols and haloalkanes, and considers the importance of polarity and bond enthalpy to organic reactions.

Throughout this section, there are many opportunities for developing organic practical skills, including preparation and purification of organic liquids.

Finally, the important techniques of infrared spectroscopy and mass spectrometry are used to illustrate instrumental analysis as a valuable tool for identifying organic compounds.

4.2.1 Alcohols

Learning outcomes	Additional guidance
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Learners should be able to demonstrate and apply their knowledge and understanding of:

Properties of alcohols

- (a) (i) the polarity of alcohols and an explanation, in terms of hydrogen bonding, of the water solubility and the relatively low volatility of alcohols compared with alkanes (**see also**